[RFR 2027 Electronics] New Member Project Design Report

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1. Revision Log

Editor	Date	Revision
Rebekah Han	2026-10-16	1

2. Purpose

To build a temperature sensor, taking analog input and then converting it into digital to output them over the I2C bus.

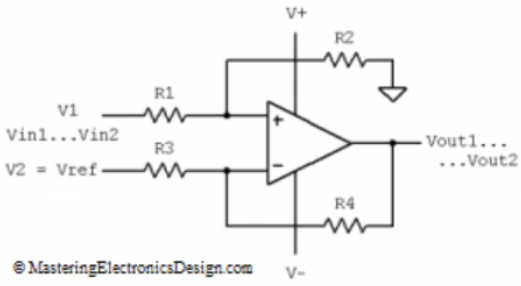
3. Parties Involved

Name	Team Role	Report Role
Rebekah Han	New Member	Author
Hank Habashy	New Member	
Shiven Jariwala	New Member	

4. Research

Given that our temperature sensors' output voltage ranges from 1.30 V to 2.44 V and that the Arduino's ADC has 10-bit resolution with a default reference voltage of 5 V, we can shift and amplify the signals from the temperature sensors in such a way that its range becomes 0 - 5 V. Keeping this in mind, we can also use an RC low-pass filter to keep lower frequency ranges since temperature signals change slowly in comparison to other signals. This will help filter out any electrical noise. To handle amplification, a differential op-amp can be used to handle both level shifting and amplification. We would like the lowest temperature (i.e. -40°C) to be 5 V instead of the original 2.44 V (from the diode) and the highest temperature (120°C) to be 0 V instead of 1.30 V. Thus the gain for the op-amp shall be about 4.38. This is because 5 V/1.14 V = 4.386, where the 5 V is from the Arduino and the 1.14 V is the range of the temperature diodes (2.44 - 1.30 = 1.14). We used this differential amplifier calculator to help us build the op-amp circuit. The image below shows the circuit and the values to use.

The multiplexer will handle from what diode we will read from at a time. The 16:1 multiplexer has 4 select pins, which allow for addressing in a way. It is helpful to think of the 4 pins corresponding to a number in binary (e.g. 0000 = 0, 0001 = 1, 0010 = 2, etc). Each number corresponds to one input pin. So if the select pins were set to 0000, then the output of the multiplexer would return data from I_0 or input "pin" 0. We can take advantage of this addressing



Given Input Range

Vin1 =	1.30 V	Vin2 =	2.44 V

Required Output Range

Vout1 =	0 V	Vout2 =	5 V

Choose a Reference Voltage

V2 =	1.3 V

Choose R2 and R3

R2 =	10.00 kOhm	R3 =	10.00 kOhm

Calculate R1 and R4

R1 =	2.28 kOhm	R4 =	43.86 kOhm

Figure 1: Op-amp calculation

with the Arduino Nano. From the Nano, we can send 1s or 0s to the 4 select pins of the multiplexer. A program can be written to cycle through all of the temperature diodes from 0 (0000) through 15 (1111) and grab the output every time a new set of bits is sent to the multiplexer. From there calculating the highest temperature, average temperature, and other analysis can be done.

5. Findings

Parts

Given Parts

- 16:1 multiplexer CD74HC4067
- Arduino Nano
- 12 to 5 V DCDC
- 16 temperature-sensitive diodes
- Red LED
- Breadboard
- Jumper cables
- Resistors

Additional Parts

The following lists parts not listed in the writeup, but are a part of our design: - Op-amp LM741 - Capacitor

Schematic

Schematic Flow Chart

Code

6. Conclusion